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A snake robot propelling inside of a pipe with helical rolling motion

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***Abstract*—Recently, multiple locomotion gaits of snakes have been realized by snake robots. A snake robot would be applied to going into a narrow space to investigate the inside of a long and slim pipe and to executing some simple work by using its slim body. In this paper, we achieve a new locomotion mode that propels a snake robot the inside of a pipe with helical rolling motion, in addition to previously achieved locomotion modes. A structure of the snake robot is constructed by connecting many units in series. The unit has a pitch axis, a yaw axis and two passive wheels. A shape of the robot is calculated by using mathematical continuum curve model. In addition, we present a method that the shape of snake robot is adjusted to a diameter of a pipe automatically. Moreover, we indicate a possibility of a new locomotion mode which conduct helical rolling motion with conical spiral form in order to improve a position of a camera while the snake robot investigates the inside of a pipe.**

***Keywords*—Snake Robot; Lateral Rolling; Helical Rolling Motion; Helical Form; Conical Spiral Form;.**

I. Introduction

A snake can achieve various functions although it is a simple form. For example, a snake can perform functions as a leg when it travels on a ground, as a hand when it manipulates something and as an arm when it coils to an object. If such snake's mechanisms are applied to engineering fields and robotics fields, it would have an ability to be a structure of multiple functional mechanical system[1]. For example, a snake robot would go into a narrow space to investigate the inside of a complicated structure and make some simple work by using its slim body[2].

There are many studies of snake robots and snake-like robots. Especially, Prof. Hirose and his colleagues are developing ACM (Active Cord Mechanism) as a snake robot[1], [3]. In the study, it is shown that a difference of friction between tangent direction and normal direction propels a snake robot by using undulatory locomotion on ground, and its body shape is called serpenoid curve. To realize undulatory locomotion by a snake robot, passive wheels are equipped with the robot so that it generates a difference of friction. The undulatory locomotion with passive wheels is very smooth and efficient locomotion. In addition to the undulatory locomotion, several locomotion gaits such as sidewinding locomotion, inch worm locomotion and lateral rolling locomotion are realized until now[3], [4]. Moreover, even a snake robot that swimming in water is developed[5].

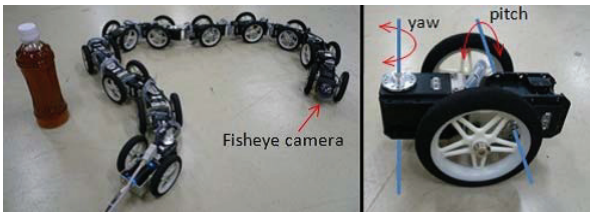

Fig. 1. Overview of the experimental snake robot and its one unit

However, previous locomotion gaits are mainly limited on two dimensional plane. A real snake can move on three dimensinal space such as climbing a tree. In the previous study, we made a snake robot with passive wheel and realized previously presented locomotion gaits such as undulatory locomotion and lateral rolling locomotion and also realized cylinder climbing locomotion by extending lateral rolling motion[6]. In the cylinder climbing locomotion, we introduce helical form to a shape of snake robot. The helical form is achieved by considering mathmatical continuum model first, and then it is applied to the mechanical discrete snake robot model [7], [8], [9], [10].

In this paper, we achieve new locomotion modes that propel the snake robot the inside of a pipe with helical rolling motion in addition to the abovementioned three locomotion modes. In the new locomotion mode, the shape of the snake robot is adjusted to a diameter of a pipe automatically. In the literature[11][12], snake robots moving in a pipe with helical rolling motion is also shown. We achieved helical rolling motion by using a snake robot equipped passive wheels. In addition, we indicate possibility of a new locomotion mode which conduct helical rolling motion with conical spiral form in order to improve a position of a camera while the snake robot investigates the inside of a pipe.

II. Structure of snake robot

The snake robot used in this study is shown in Fig. 1. The snake robot is constructed by connecting many units in series. One unit is constructed with two degree of freedom (a pitch axis and a yaw axis). It is possible to equip passive wheels with each unit. The snake robot is able to realize undulatory

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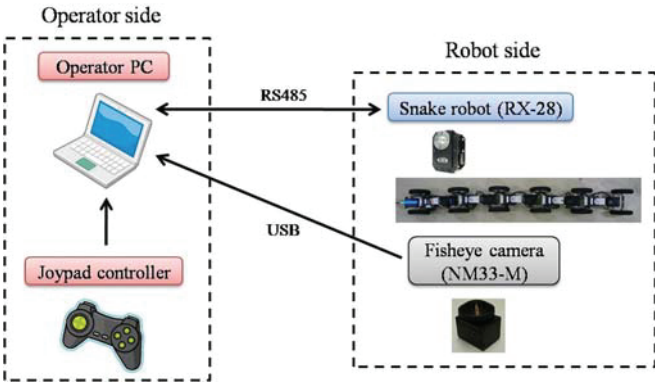

Fig. 2. System structure of experimental snake robot

TABLE I
SPECIFICATION OF RX-28

Waight [g]	72	
Dimension [mm]	35.6×50.6×35.5	
Gear Reduction Ratio	1/193	
Applied Voltage [V]	12	16
Max Holding Torque [Nm]	2.77	3.69
Max Angular Velocity [rad/s]	6.27	8.31
Feedback	Position, Velocity, Temperature, Torque, Input Voltage	

locomotion due to equipped passive wheels. The length of one unit is 145[mm], the width and height is 50[mm],the tread of wheels is 88[mm] and the weight is 210[g].

The system configuration of the snake robot is shown in Fig. 2. As an actuator of the robot, we adopt Dynamixel RX-28. The specification of RX-28 is shown in Table I. RX-28 has its control circuits inside of its body, and each joint angle is controlled by sending specific command to it using serial RS485 communication. The outside of passive wheel is made of sponge. The diameter of the wheel is 74[mm]. Also, the stabilizing power supply is connected to the snake robot as external electrical power source. The communication line and the power supply line are connected with the robot using a slip ring at the end of the robot. The snake robot is assumed to enter a pipe and to inspect inside of it. Therefore, the fisheye camera (OPT Corporation：NM33-M) is equipped at the head of snake robot. NM33-M has a fisheye lens which provide a 214[deg] field of view.

III. CONSIDERATION FOR METHODS OF PROPELLING THE INSIDE OF A PIPE

There are various kinds of pipes. For instance, there are vertical pipes, horizontal pipes, elbow pipes, T-pipes, and pipes changing diameter on the way. Therefore, it is desirable that a robot is able to move in the various pipe by adjusting its shape and locomotion strategy to these pipes. It is necessary that a robot exerts perpendicular force at the inner wall of a pipe appropriately when it moves the inside of a pipe. A snake robot is a redundancy system and long and slender. Then, it is thought that a snake robot can adjust to various pipes by changing its shape.

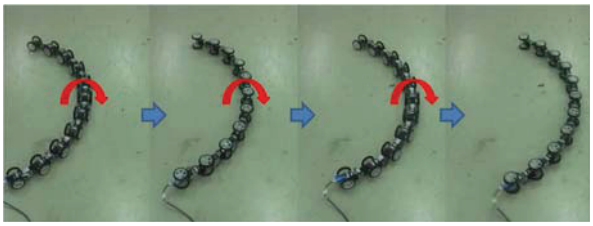

Fig. 3. Lateral rolling locomotion

The following two locomotion strategies are thought as a method that a snake robot exerts perpendicular force at the inner wall of a pipe when it moves the inside of a pipe.

1) Method of locomotion by driving body trunk like sine wave[13]. (Sinusoidal wave drive principle)
2) Method of locomotion by propelling with helical rolling motion. (Helical rolling motion)

A snake robot that moves by sinusoidal wave drive principle causes a few contact points to a inner wall of pipe. It is thought that the robot moving by the method in vertical pipe requires large power and friction force against for the weight of the robot. On the other hand, the robot that moves by helical rolling motion causes many contact points to a inner wall of pipe. It is considered that the snake robot moving by helical rolling motion requires smaller power and friction force than sinusoidal wave driving principle. Therefore, the locomotion strategy by helical rolling motion has an advantage.

There is a locomotion mode for a snake robot named lateral rolling locomotion moving on a two dimensional plane. The appearance of lateral rolling locomotion is shown in Fig. 3. Lateral rolling locomotion is achieved by twisting a snake robot body without a roll axis, as a result the robot moves lateral direction to the robot. The rolling motion is generated by giving the pitch axis and the yaw axis a sine wave that has phase shift of 90[deg]. Desired angle for each joint is given as follow

$$ {}^{p}\theta^{d} = A\sin(\omega t) \tag{1} $$

$$ {}^{y}\theta^{d} = A\cos(\omega t) \tag{2} $$

where, ${}^{p}\theta^{d}$ is a desired angle for all pitch joints, ${}^{y}\theta^{d}$ is a desired angle for all yaw joints, A is an amplitude, ω is an angular frequency and t is a time. The body trunk of the snake robot is shaped into the character of ”U” by giving the amplitude A, and generates rolling motion.

Helical rolling motion in three dimensional space is generated by expanding the lateral rolling locomotion. Shape creation of helical form in three dimensional space is described below.

IV. SHAPE CREATION OF A SNAKE ROBOT

To express a snake robot shape, a continuous curve model is used. First of all, the shape of a snake robot is expressed by using a continuous curve model of Frenet-Serret formula in

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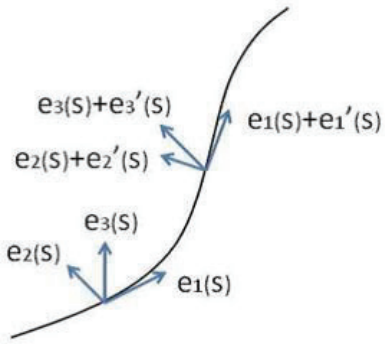

Fig. 4. Coordination system of Serret-Frenet

three dimensional space. Afterward, it is applied to the discrete line model.

A. Mathematical continuum model

Generally speaking, continuum curve in three dimensional space is expressed by equation of Serret-Frenet, which can express the curve $\mathbf{c}(s)$ by using curvature $\kappa(s)$ and torsion $\tau(s)$.

$$\begin{aligned} d\mathbf{c}(s)/ds &= \mathbf{e_1}(s) \\ d\mathbf{e_1}(s)/ds &= \kappa(s)\mathbf{e_2}(s) \\ d\mathbf{e_2}(s)/ds &= -\kappa(s)\mathbf{e_1}(s) + \tau(s)\mathbf{e_3}(s) \\ d\mathbf{e_3}(s)/ds &= -\tau(s)\mathbf{e_2}(s) \end{aligned} \tag{3}$$

where, s is a parameter of length along the curve, $\mathbf{e_1}(s)$, $\mathbf{e_2}(s)$, $\mathbf{e_3}(s)$ are orthonormal basis unit vector. $\mathbf{e_1}(s)$ is a tangential vector to the curve at s, $\mathbf{e_2}(s)$ is a vector that indicates the direction of changing the curve at s and $\mathbf{e_3}(s)$ is a vector that is given by $\mathbf{e_3}(s) = \mathbf{e_1}(s) \times \mathbf{e_2}(s)$ as shown in Fig. 4.

The formulation of Serret-Frenet equation is not suitable for the robot coordinate system, because it gives only a shape of curve, it does not give a concept corresponds to a joint structure of a robot. So, a coordinate system for a mechanical snake robot $\mathbf{e_r}(s)$, $\mathbf{e_p}(s)$, $\mathbf{e_y}(s)$ is derived[7] as shown in Fig. 5. Let $\tau_{roll}(s)$, $\kappa_{pitch}(s)$, $\kappa_{yaw}(s)$ be a torsion of roll, curvature of pitch and curvature of yaw at point s on the continuum robot coordinate system respectively. In this formulation, the continuum curve is given by following.

$$\begin{aligned} d\mathbf{c}(s)/ds &= \mathbf{e_r}(s) \\ d\mathbf{e_r}(s)/ds &= \kappa_{yaw}(s)\mathbf{e_p}(s) - \kappa_{pitch}(s)\mathbf{e_y}(s) \\ d\mathbf{e_p}(s)/ds &= -\kappa_{yaw}(s)\mathbf{e_r}(s) + \tau_{roll}(s)\mathbf{e_y}(s) \\ d\mathbf{e_y}(s)/ds &= \kappa_{pitch}(s)\mathbf{e_r}(s) - \tau_{roll}(s)\mathbf{e_p}(s) \end{aligned} \tag{4}$$

Finally, by using this formula, $\tau_{roll}(s)$, $\kappa_{pitch}(s)$, $\kappa_{yaw}(s)$ decide the shape of continuum curve in three-dimensional space.

Let us recall that the snake robot in this paper has only pitch axes and yaw axes. There is no roll axis with the snake robot. Because of this construction, $\tau_{roll}(s)$ in the eq. (4) become $\tau_{roll}(s) = 0$. So eq. (4) to the snake robot in this paper becomes as follow.

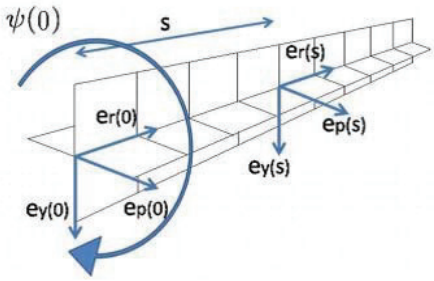

Fig. 5. Coordination system of a snake robot

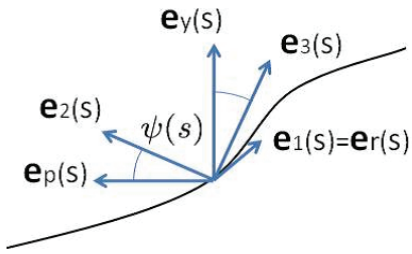

Fig. 6. $\psi(s)$: difference of angle between $\mathbf{e_2}(s)$ and $\mathbf{e_p}(s)$

$$\begin{aligned} d\mathbf{c}(s)/ds &= \mathbf{e_r}(s) \\ d\mathbf{e_r}(s)/ds &= \kappa_{yaw}(s)\mathbf{e_p}(s) - \kappa_{pitch}(s)\mathbf{e_y}(s) \\ d\mathbf{e_p}(s)/ds &= -\kappa_{yaw}(s)\mathbf{e_r}(s) \\ d\mathbf{e_y}(s)/ds &= \kappa_{pitch}(s)\mathbf{e_r}(s) \end{aligned} \tag{5}$$

To make eq. (5) coincide with eq. (3), calculate eight formula with $\mathbf{e_1}(s) = \mathbf{e_r}(s)$, then expression of $\kappa_{pitch}(s)$, $\kappa_{yaw}(s)$ using $\kappa(s)$, $\tau(s)$ is derived as below.

$$\begin{aligned} \kappa_{pitch}(s) &= -\kappa(s)\sin(\psi(s)) \\ \kappa_{yaw}(s) &= \kappa(s)\cos(\psi(s)) \\ \psi(s) &= \int_0^s \tau(s)ds + \psi(0) \end{aligned} \tag{6}$$

where, $\psi(s)$ is a difference of angle between $\mathbf{e_2}(s)$ and $\mathbf{e_p}(s)$ as shown in Fig. 6. $\psi(0)$ is an integral constant value, it can be decided by user freely.

Helical form is realized to set parameters $\kappa(s)$ and $\tau(s)$ as a certain constant value in eq. (6). Formula expressing of helical form becomes as below.

$$\begin{aligned} \begin{bmatrix} \kappa_{pitch}(s) \\ \kappa_{yaw}(s) \end{bmatrix} &= \begin{bmatrix} -\kappa\sin(\psi(s)) \\ \kappa\cos(\psi(s)) \end{bmatrix} \\ \psi(s) &= \int_0^s \tau ds + \psi(0) \end{aligned} \tag{7}$$

By changeing the value of κ and τ in eq. (7), a diameter and a pitch of helical form to be changed respectively. In addition, by changing the value of $\psi(0)$, the continuum robot coordinate

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system is rolled around $\mathbf{e_r}(s)$ so that the robot achieve lateral rolling motion.

B. Apply to discrete mechanical snake robot model

In the case that a length between pitch joint and yaw joint is δs, the amount of rotational value for each pitch joint and yaw joint of mechanical snake robot can be expressed as $\kappa_{pitch}(s)\,ds$ and $\kappa_{yaw}(s)\,ds$ respectively. Therefore, target angle of i-th joint is expressed as below.

$$\theta_i^d = \begin{cases} 2\delta s \kappa_{pitch}(i\delta s) & (i : \text{oddjoint}) \\ 2\delta s \kappa_{yaw}(i\delta s) & (i : \text{evenjoint}) \end{cases} \tag{8}$$

In this regard, odd number joint is pitch joint and even number joint is yaw joint. The coefficient "2" represents that interval of each pitch joints or yaw joints is $2\delta s$.

Note that for the mechanical snake robot model in this paper, rotation angle of each roll axis of the unit can not be controlled. Therefore presented helical form generates a gap to the roll axis of the robot. The gap is integrated along s as the $\psi(s)$. As a result, each passive wheel does not touch at the same face to inner wall of a pipe.

C. Formulation of helical form and conical spiral form

An arbitrary curve can be generated by giving parameters of curvature $\kappa(s)$ and torsion $\tau(s)$ in eq. (6). The parameters of the curvature and the torsion to achieve helical form and conical spiral form are shown as follows.

1) In the case of helical form: In the cartesian reference curve, the polar representation $\mathbf{f}(t) = (x(t), y(t), z(t))^T$ of a conical spiral curve is given as follow.

$$\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} a\cos(t) \\ a\sin(t) \\ bt \end{pmatrix} \tag{9}$$

where, t is a mediation parameter. a is a helical curve radius. b is an increasing rate to z-axis. The variable parameter of the curvature and the torsion for a conical spiral curve are derived from the eq. (9) as follows.

$$\kappa(s) = a/\left(a^2+b^2\right) \quad , \quad \tau(s) = b/\left(a^2+b^2\right) \tag{10}$$

In the case of helical curve, curvature $\kappa(s)$ and torsion $\tau(s)$ are constants regardless of the value s that denote length alog the curve.

2) In the case of conical spiral form: In the cartesian reference curve, the polar representation $\mathbf{f}(t) = (x(t), y(t), z(t))^T$ of a conical spiral curve is given as follow.

$$\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} at\cos(t) \\ at\sin(t) \\ bt \end{pmatrix} \tag{11}$$

where, t is a mediation parameter. a is an increasing rate of spiral radius. b is an increasing rate to z-axis. The variable parameter of the curvature and the torsion for a conical spiral curve are derived from the eq. (11) as follow.

$$\begin{aligned} \kappa(s) &= \frac{a\sqrt{a^2\left(t^2+2\right)^2 + b^2\left(t^2+4\right)}}{\left\{a^2\left(t^2+1\right)+b^2\right\}^{\frac{3}{2}}} \\ \tau(s) &= \frac{b\left(t^2+6\right)}{a^2\left(t^2+2\right)^2 + b^2\left(t^2+4\right)} \end{aligned} \tag{12}$$

In the case of conical spiral curve, curvature $\kappa(s)$ and torsion $\tau(s)$ are variables for the value s because it changes according to the parameter t. Note that, relation between length s along the curve and mediation parameter t is given as follow.

$$\frac{ds}{dt} = \left\| \frac{df(t)}{dt} \right\| = \sqrt{\left(\frac{df(t)}{dt}\right)^2} \tag{13}$$

D. Adaptation algorithm

It is desirable that the snake robot exerts perpendicular force at the inner wall of a pipe appropriately when it moves inside of a pipe. So, we introduce adaptation algorithm to the snake robot. First, the snake robot is set its helical curve radius smaller than the diameter of the pipe. Then, the helical curve radius is enlarged gradually. The helical curve radius is controlled according to the given threshold value. The outline of the adaptation algorithm that changes the helical curve radius is shown below.

- Step1 Measure a load of actuator every sampling time and calculate the mean value for a certain term.
- Step2 If the mean value is smaller than the given threshold value, the helical curve radius is enlarged, otherwise the helical curve radius is reduced.
- Step3 Go to Step1.

Actually, the value of the load feedbacked from RX-28 is averaged and used as a value to decide the helical curve radius. The feedbacked load value from RX-28 is obtained as 10bit data by changing torque by AD conversion. In the cass the power-supply voltage for RX-28 is 16[V], the AD value of the load from 0 to 1023 converted into the value of the torque from 0 to 3.69[Nm] (see Table I). The adaptation algorithm is only applied to the case of helical form because the load value from RX-28 is available only one joint in the current version of experimental system. A joint located center of the snake robot is chosen as the measuring joint.

V. Experimental results of propelling inside of a pipe with helical form

A. Objective of Experiment

In this section, experimental results propelling a snake robot the inside of a pipe with helical form are shown. Following are objectives of experiments. First aim is to confirm whether it is possible to move inside of a horizontal and vertical pipe with helical form by using helical rolling motion. Second aim is to confirm whether it is possible to achieve autonomously-changing of helical shape and to move by adjusting to a diameter of a pipe using adaptation algorithm. Third aim is to observe the image of fisheye camera NM33-M that mounted

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TABLE II
SPECIFICATION AND PARAMETERS OF SNAKE ROBOT USED FOR EXPERIMENTS

Total mass [g]	2550
Total length [mm]	1540
Number of units	10
Applied Voltage [V]	16
Twisting motion $\dot{\psi}(0)$ [rad/s]	4.0

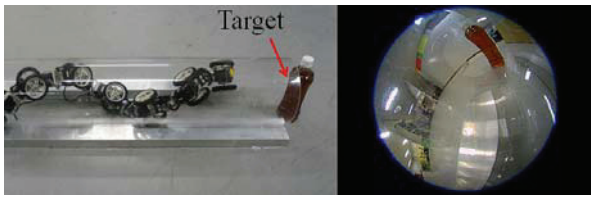

Fig. 7. An experimental result of propelling in a pipe with helical form (for the pipe inner diameter 195[mm])

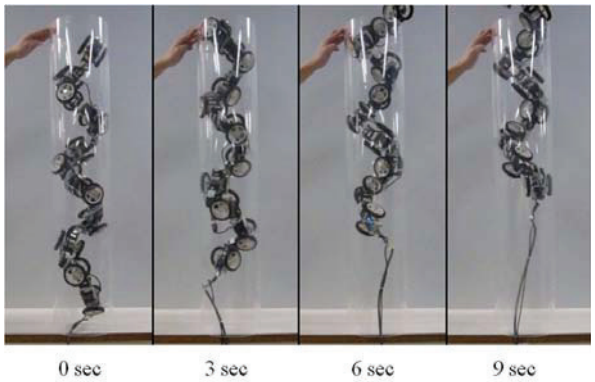

Fig. 8. An experimental result of propelling in a vertical pipe with helical form (for the pipe inner diameter 195[mm])

on the first unit, and to ascertain whether it is useful for the inspection of the inside of a pipe. The specification and parameters of the snake robot used for experiments is shown in Table II.

B. Experimental result

1) In the case of propelling the inside of a horizontal pipe: In this experiment, we prepare two type of experimental pipes made of acryl, each inner diameter is 145[mm] and 195[mm]. An example of experimental result propelling the inside of horizontal pipe inner diameter 195[mm] is shown in left side of Fig. 7.

For the pipe inner diameter 145[mm], the parameter of helical form is fixed as κ is 0.77 and τ is 1.25 . As a result, the robot is propelled with velocity of 33.1[mm/s]. For the pipe inner diameter 195[mm], the parameter of helical form is fixed as κ is 0.95 and τ is 0.65. As a result, the robot is propelled with velocity of 66.4[mm/s].

2) In the case of propelling the inside of a vertical pipe: In this experiment, the same two type of pipes are used. An example of experimental result propelling the inside of vertical

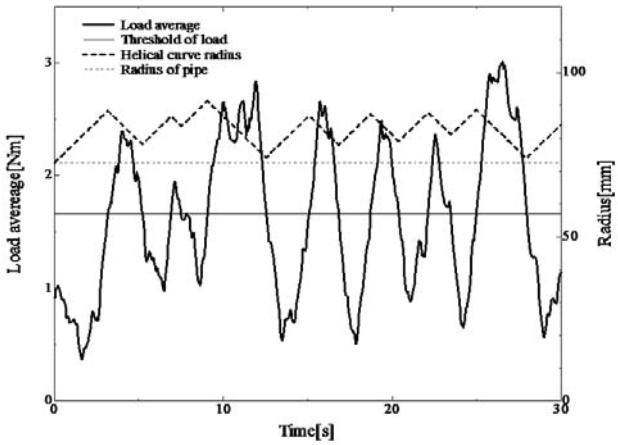

Fig. 9. An experimental result of adaptive algorithm with helical form (for the pipe inner diameter 145[mm])

pipe inner diameter 195[mm] is shown in Fig. 8.

For the pipe inner diameter 145[mm], the parameter of helical form is also fixed as κ is 0.77 and τ is 1.25. As a result, the robot is propelled with velocity of 9.49[mm/s]. Each propelling speed, decrease comparing with horizontal pipe due to a slip of contact points. For the pipe inner diameter 195[mm], the parameter of helical form is also fixed as κ is 0.95 and τ is 0.65. As a result, the robot is propelled with velocity of 48.0[mm/s].

From those experiments, it is confirmed that helical rolling motion can achieve propelling the snake robot in both hotizonal and vertical pipe for different diameters.

3) In the case of propelling the inside of a horizontal pipe with adaption algorithm: In this experiment, the same two type of pipes are also used. The graph of Fig. 9 shows relations between the average of load feedbacked from RX-28 and helical curve radius while the robot is adaptively moving in the horizontal pipe inner diameter 145[mm] using adaptation algorithm. In this experiment, the load value is averaged by 150 sampling value. The increace/decreace ratio of a radius is 0.05[mm] per 1 step (10[ms]). As a result, the robot autonomously adapted to inner diameter of pipe. The slight continuous line in Fig. 9 shows threshold value (threshold value is set as 1.66[Nm]) for adaptation algorithm, and the slight dash line shows the radius of pipe. From Fig. 9, helical radius (which is added robot width from center to wheel 40[mm]) proportionally increase while the lode average is below the threshold value, and proportionally decrease while the lode average is above the threshold value. This is repeating, and helical radius fluctuates around approximate 80[mm]. Note that the helical radius in Fig. 9 becomes bigger than the radius of pipe 72.5[mm]. The reason of this result is caused that exerting perpendicular force to the inner wall of pipe is due to a effect of servo of RX-28. In order to add perpendicular force to the wall, the helical radius must bigger than the radius of pipe, namely, the helical radius is not actual value but desired

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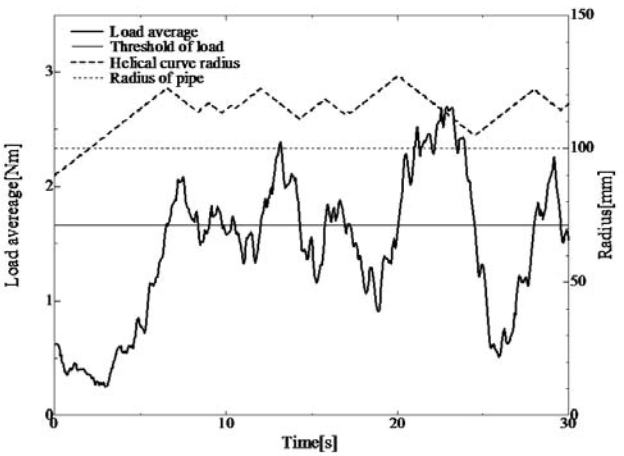

Fig. 10. An experimental result of adaptive algorithm with helical form (for the pipe inner diameter 195[mm])

Fig. 11. Overview of the conical spiral form

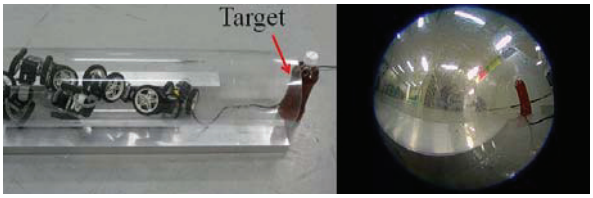

Fig. 12. An experimental result of propelling in a pipe with conical spiral form (for the pipe inner diameter 195[mm])

value. Moving velocity of this experiment is 24.2[mm/s] .

The graph of Fig. 10 shows an experimental result for horizontal pipe inner diameter 195[mm] using adaptation algorithm. The threshold value for adaptation algorithm is also 1.66[Nm]. Behavior of the robot is the same as mentioned above. The helical radius fluctuates around approximate 120[mm]. The helical radius in Fig. 10 becomes bigger than the radius of pipe 97.5[mm]. This is caused by the same reason as mentioned above. Moving velocity of this experiment is 40.2[mm/s]. From those experiments, it is confirmed that the snake robot can achieve adaptive helical form according to a diameter of pipe.

4) Position of fisheye camera: The right side of Fig. 7 shows the fisheye camera image for the experiment. It is confirmed that the snake robot is able to move in the pipe by proposed method. However, by this form, fisheye camera image becomes a biased image because the fisheye camera goes along to the inner wall and the target never can be caught at the center of the image. Therefore, helical form may not be so suitable for a inspectional task.

VI. Experimental results of propelling inside of a pipe with conical spiral form

A. Objective of Experiment

In this section, experimental results propelling a snake robot the inside of a pipe with conical spiral form are shown. Following are objectives of experiments. First aim is to confirm whether it is possible to move inside of a horizontal pipe with helical form by using helical rolling motion. Second aim is to observe the image of fisheye camera, and to confirm whether it is improved to the helical form. The specification and parameters of the snake robot used to experiments is the same as previous experiments shown in Table II.

B. Experimental result

The conical spiral form is shown in Fig. 11. As shown Fig. 11, the radius of conical spiral increases gradually.

1) Propelling the inside of a horizontal pipe: In this experiments, the pipe inner diameter 195[mm] is used. An example of experimental result propelling inside of horizontal pipe by conical spiral form with helical rolling motion is shown in left side of Fig. 12. The parameter of conical spiral form is fixed as a is 4[mm] and b is 25[mm]. As a result, the robot is prpoelled with velocity of 46.4[mm/s].

2) Position of fisheye camera: The right side of Fig. 12 shows the fisheye camera image for the experiment. Comparing with helical form, the position of camera is located almost center of the pipe. Therefore, conical spiral form is preferable to helical form for a inspectional task. However, camera still does not look to the target but to the wall of the pipe. To improve this characteristic, several units from first unit may need another control method.

VII. Conclusion and future work

In this study, we proposed the method to propel the snake robot inside of a pipe with helical rolling motion. Moreover, the adaptable algorithm that adjust helical curve radius to a diameter of the pipe was presented. In addition, we proposed a new form that propel the snake robot in a pipe by helical form and conical spiral form by considering a camera position. These proposal are confirmed by experimental results.

Future work will devoted to apply adaptation algorithm to conical spiral form and to achieve investigation in the winding pipe.

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